

HOW VISION COMPUTING SoCs Will Reshape India's Electronics And Embedded Systems Landscape

Vision Computing SoCs could emerge as a defining layer in India's next phase of electronics growth. With the right ecosystem in place, the country can move beyond following global hardware trends and begin shaping the future of intelligent embedded systems.



patterns, generate insights, and transmit information in real time.

Vision Computing SoCs (system-on-chip) are becoming the quiet engine behind this shift. These specialised chips do far more than process data. They decode images, analyse surroundings, and trigger real-time responses—often within fractions of a second and without relying on remote servers. From autonomous drones operating above agricultural fields to next-generation surveillance systems monitoring urban environments, Vision Computing SoCs integrate AI, neural networks, and sensor capabilities into a single efficient platform.

Why does this matter now? As India moves towards a \$300 billion electronics production target by 2026, the ability to build smarter and more energy-efficient systems has become essential. Vision Computing SoCs could play a pivotal role in bridging India's ambition with its expanding innovation capability.

India's old semiconductor playbook is not enough

Even as India sets its sights on becoming a global electronics hub, several structural roadblocks remain. One of the most significant challenges is the country's continued dependence on imported semiconductors. With India's chip demand projected to grow from \$54 billion in 2025 to \$108 billion by 2030, this reliance not only increases costs but also exposes manufacturers to supply chain risks.

Despite government-backed initiatives and new investments in semiconductor manufacturing, the industry remains in its early stages.

Semiconductor chips have long been the backbone of advances in emerging technologies such as AI, AR, VR, and related domains. Unlocking the full potential of these technologies increasingly depends on specialised chips designed for specific workloads.

Much like IoT, vision computing is emerging as a vital capability for connected devices. While IoT enables constant connectivity and data transfer, vision computing goes a step further by extracting actionable insights from that data. Vision-enabled devices are expected not only to "see" but also to identify



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